

1/2/2026

Model Lab Practical

classmate

Date _____

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① Exp : 1

Aim:

To design and configure a network using Cisco packet tracer to demonstrate basic load balancing across two servers.

Procedure:

- ① open cisco packet Tracer and place two servers, a switch, a Router, and multiple End devices (PCs)
- ② connect the PCs and servers to the switch and connect the switch to the Router.
- ③ Access the Router's CLI or config tab. Configure the interface connected to the switch with an IP address and enable the port using the command.
- ④ Assign static IP addresses to server 0 & server 1. Set their gateways.
- ⑤ configure the PCs with IP addresses in the same subnet.
- ⑥ To simulate load distribution, add a DNS server to the network & create two separate 'A records' for the exact same domain name.

⑦ Update the DNS settings on all PCs to point to the DNS server's IP address

⑧ From different PCs open the command prompt and ping to observe the DNS round-trip load distribution in action

Sample input and output:

- input (PC 1).

ping [www.data-center.com]

- output:

Reply from 192.168.1.10.

- input (PC 2)

ping [www.data-center.com]

- output (PC 2)

Reply from 192.168.1.11.

Result:

The network was successfully designed and configured in packet tracer, and

basic load balancing across two servers was demonstrated using DNS round-robin distribution

Exp 2:

Aim:

To configure port security on a switch in Packet Tracer to allow only specific MAC addresses to connect to a port and to verify that unauthorized devices are successfully blocked.

Procedure:-

- ① Add a switch, one authorized PC (PC-A), and one unauthorized PC (PC-B) to the packet tracer workspace
- ② Connect PC-A to the switch's Fast Ethernet 0/1 port using a copper straight-through cable. Assign IP addresses to both PCs in the same subnet.
192.168.1.2 a) 192.168.1.3
- ③ Access the Switch CLI, enter global configuration mode, and select the interface.
- ④ Force the port into access mode.
- ⑤ Enable the port security feature.
- ⑥ Restrict the port to allow only a

single mac address

⑦ configure the switch to dynamically learn and lock the authorized device connects.

⑧ Set the security violation mode to shut down the port if an unauthorized device connects.

⑨ Ping the switch or another device from PC-A to generate traffic to the switch learns the Mac address

⑩ disconnect PC-A, connect PC-B to Fast Ethernet 0/1, and attempt to ping. Observe the port link state.

Sample input and output:

• input

ping 192.168.1.1

• output

reply from 192.168.1.1

• input :

ping 192.168.1.1.

output :

Request timed out.

Result:

Port security was successfully configured on the switch. It was verified that the authorized mac address was permitted to transmit data.

3) Wireshark analysis of traceroute packets

Aim:

To capture and analysis traceroute packets using Wireshark and study TTL & ICMP.

Procedure:

- ⇒ open a command prompt
- ⇒ start Wireshark & select the active network interface.

- ⇒ start packet capture.

- ⇒ Apply the filter.

- ⇒ Examine the captured ICMP

packet

- ⇒ Each router represent one hop between the source and destination

- ⇒ Stop the capture after the traceroute is completed.

working :-

Traceroute works by sending packets with gradually increasing TTL values. Wireshark allows these packets & their header field to be examined.

Result :-

Traceroute packets were successfully captured & analysed using Wireshark. The intermediate network hops were identified using TTL and ICMP messages.

~~C program~~

Aim :-

To write a C program to generate a parity bit & detect errors in transmitted data.

Procedure :-

- ⇒ Read the no. of data bits
- ⇒ Enter the binary data
- ⇒ Count the no. of 1s
- ⇒ Generate an even parity bit

- Add parity bit to the data.
- ⇒ count the total no. of 1s
- ⇒ If the total is even, there is no errors.
- ⇒ If the total is odd, an error is detected.

C - program:

```
#include <stdio.h>

int main () {
    int n, i, ones = 0;
    scanf ("%d", &n);
    int a[n];
    for (i = 0; i < n; i++) {
        scanf ("%d", &a[i]);
        if (a[i] == 1) ones++;
    }
    printf ("parity bit : %d\n",
           ones % 2);
    if (ones + ones % 2 == 0)
        printf ("No error")
    else
        printf ("Error detected");
    return 0;
}
```

Result:

The C program successfully generates an even parity bit and detects errors in the given binary data.